Radiation Code Analysis

Swetha Arumugam

swethaa@clemson.edu

1 Introduction

This is my final project for the graduate-level Electrodynamics course. In this project, we primarily studied the EM radiation emitted by an accelerating point charge undergoing linear oscillatory motion along $\hat{z}$ direction. Using the Liénard–Wiechert potentials, we compute the electric and magnetic fields at distant observation points and analyze the resulting radiation properties.

The code numerically solves for the retarted time t_{ret} using scipy's fsolve function, constructs E and B fields, computes the Poynting vector, and evaluates the average angular power distribution. We first checked all the results with the theoretical results we derived in the class. We then explore the behavior of the radiation for different particle velocities to observe the break in Newtonian limit, requiring $\omega l \ll c$. We also observe the onset of the radiation zone by comparing the numerical angular distribution which is proportional to $\sin^2\theta$ dependence in the far-field limit.

2 Testing

In manual calculation of t_{ret} such that it is equal to zero at $r_{\text{obs}} = [1, 0, 0]$, gives $t_{\text{obs}} = \frac{\sqrt{2}}{c}$. We used that to verify if retarded time function works well.

```
retarded time:  [0.]
observer time:  4.714045207910317e-11
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Figure 1: We see that we got the results we expected:)

We can further verify this by studying for a particle oscillating along $\hat{z}$ direction. We studied that the antenna pattern should look something like a donut when viewed along xz or yz plane. Refer fig. 2.

We can then verify that for under Newtonian limit, E and B fields are sinusoidal. The E and B fields are calculated from ϕ and $\mathbf{A}$. They can be calculated exactly (like given in textbook) or using finite differencing method. And we did in both ways and they give approximately same result if the delta for finite-differencing is chosen appropriately.

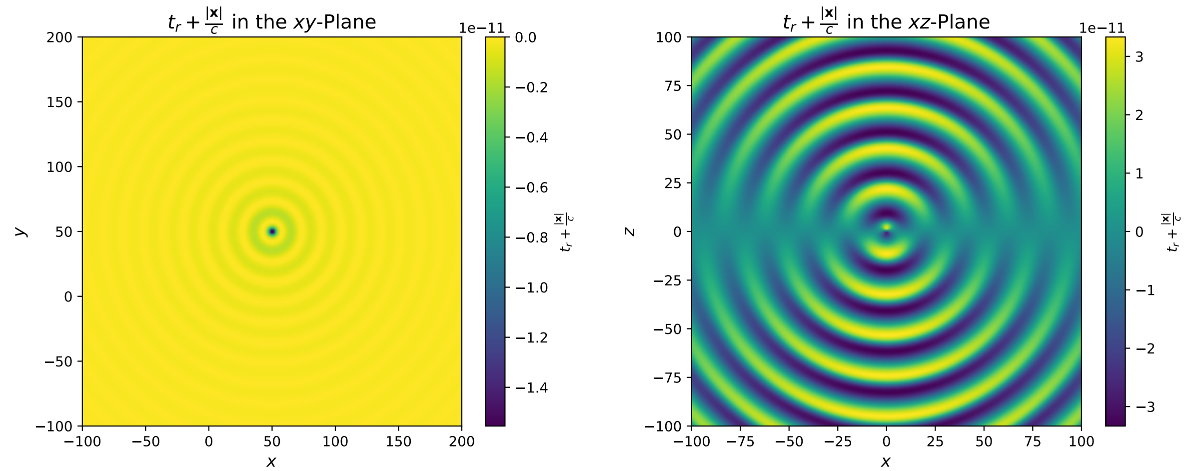


Figure 2: We can see that retarded time function is working very well. The test was done for $\beta = \frac{v}{c} = 0.3$

Question: why is B not sinusoidal? (B_y is sinusoidal but $|\mathbf{B}|$ somehow isn't. Tried resolving but it didn't work. Also, as r increases why does the initial phase of E and B field change?

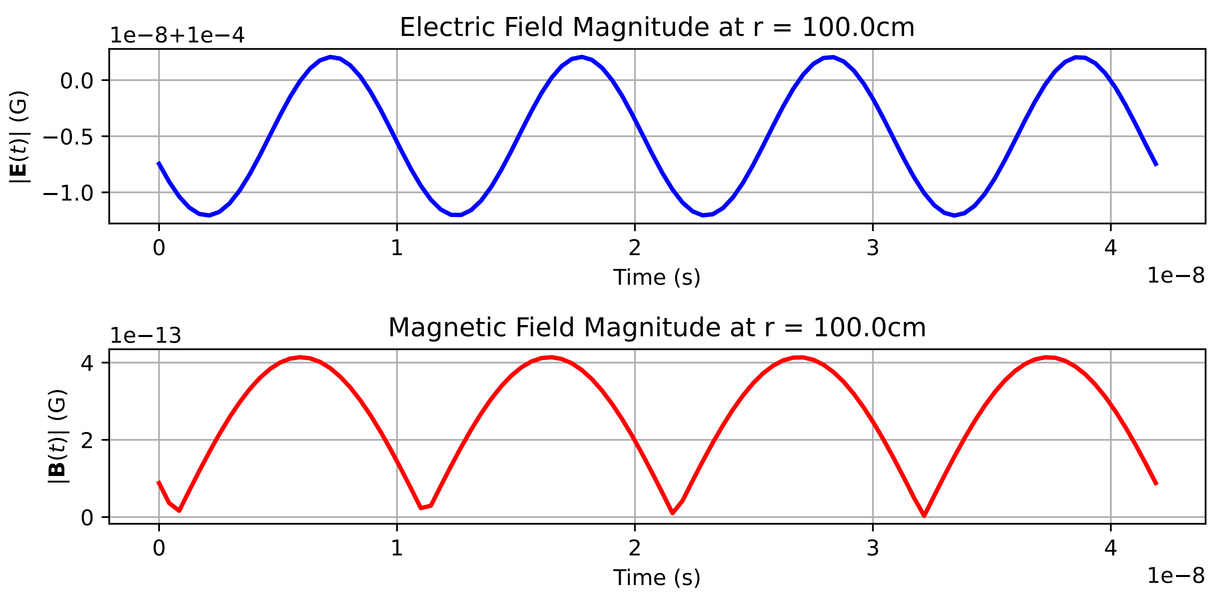
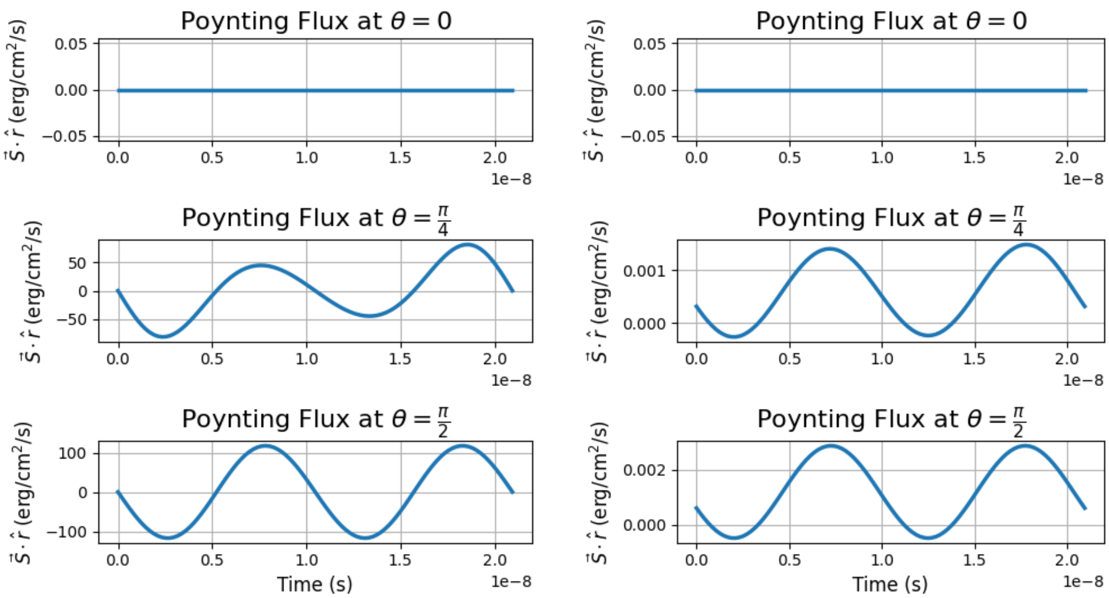


Figure 3: $|(E(t))|$ and $|(B(t))|$ for $\beta = \frac{v}{c} = 0.3$.

Fig. 4 shows the Poynting flux $\mathbf{S}(\mathbf{r}, t, \theta) \cdot \hat{\mathbf{r}}$ at different polar angles $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$, at fixed radial observation point. At $\theta = 0$, along the axis of oscillation, no energy is radiated outward - consistent with known radiation pattern (we learnt in class:)). As θ increases, the flux becomes non-zero and is oscillatory, reaching its maximum amplitude in the equatorial plane $\theta = \frac{\pi}{2}$. This behavior confirms the expected angular dependence of the radiation power, which is proportional to $\sin^2\theta$ in the Newtonian limit. Compared to r=10, the flux amplitudes for r=100 are significantly reduced, illustrating the $\frac{1}{r^2}$ falloff characteristic of energy in the radiation zone. The waveform remains periodic but shows clearer sinusoidal structure at larger distances due to the dominance of far-field terms.



(a) r = 10cm

(b) r = 100cm

Figure 4: Time Evolution of the Radial Poynting Flux $\mathbf{S}(\mathbf{r}, t, \theta) \cdot \hat{\mathbf{r}}$ at different polar angles and radial distances. This test was done for $\beta = 0.01$

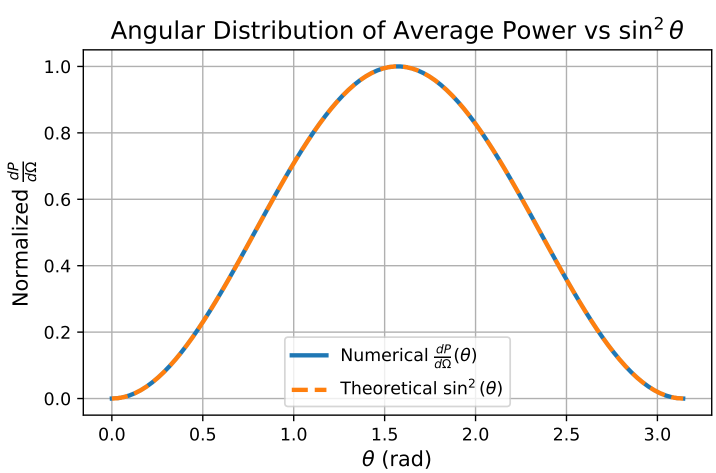


Figure 5: Compares numerically computed angular power distribution (normalized) $\frac{dP}{d\Omega}(\theta)$ at r=100cm with the theoretical distribution $\propto \sin^2\theta$.

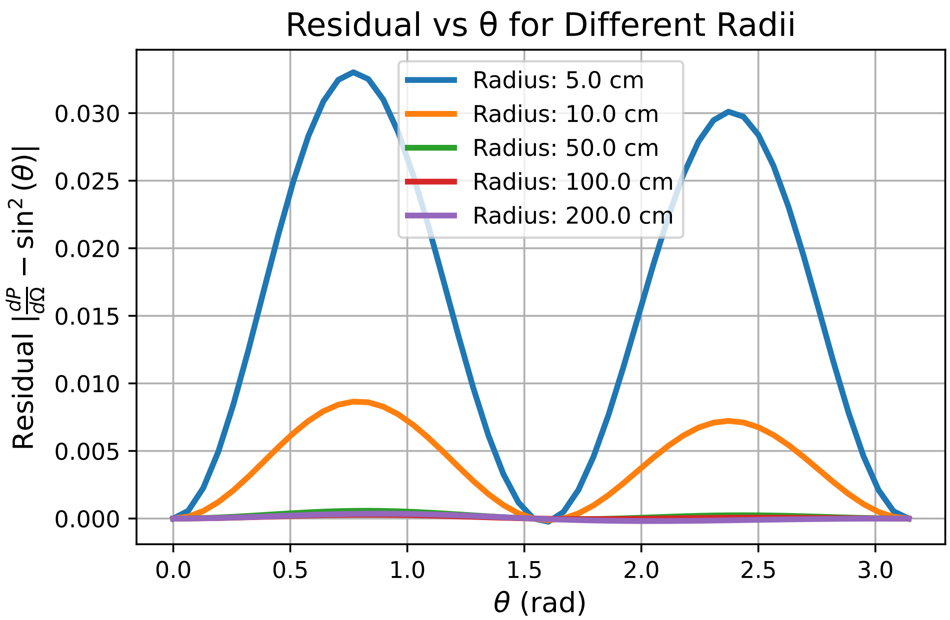


Figure 6: shows the residual between numerically computed angular power distribution (normalized) $\frac{dP}{d\Omega}(\theta)$ and the theoretical distribution $\propto \sin^2\theta$ for various observation radii. At small radii, the deviation is significant, indicating that we are in the near-field region. As the radius increases, the residual decreases, approaching zero in the far-field limit.

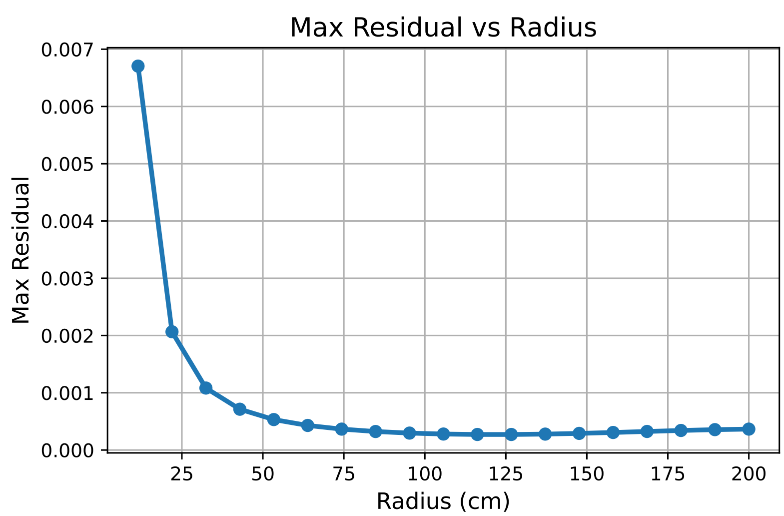


Figure 7: helps us clearly find where the far-field radiation starts to become dominant. We can see the residual decays with increasing radius, confirming convergence toward the far-field (radiation zone) behavior. Around r=100cm, the residual becomes negligible, marking the onset of the radiation regime.

Question: Why does the maximum residual increase again after 125cm?

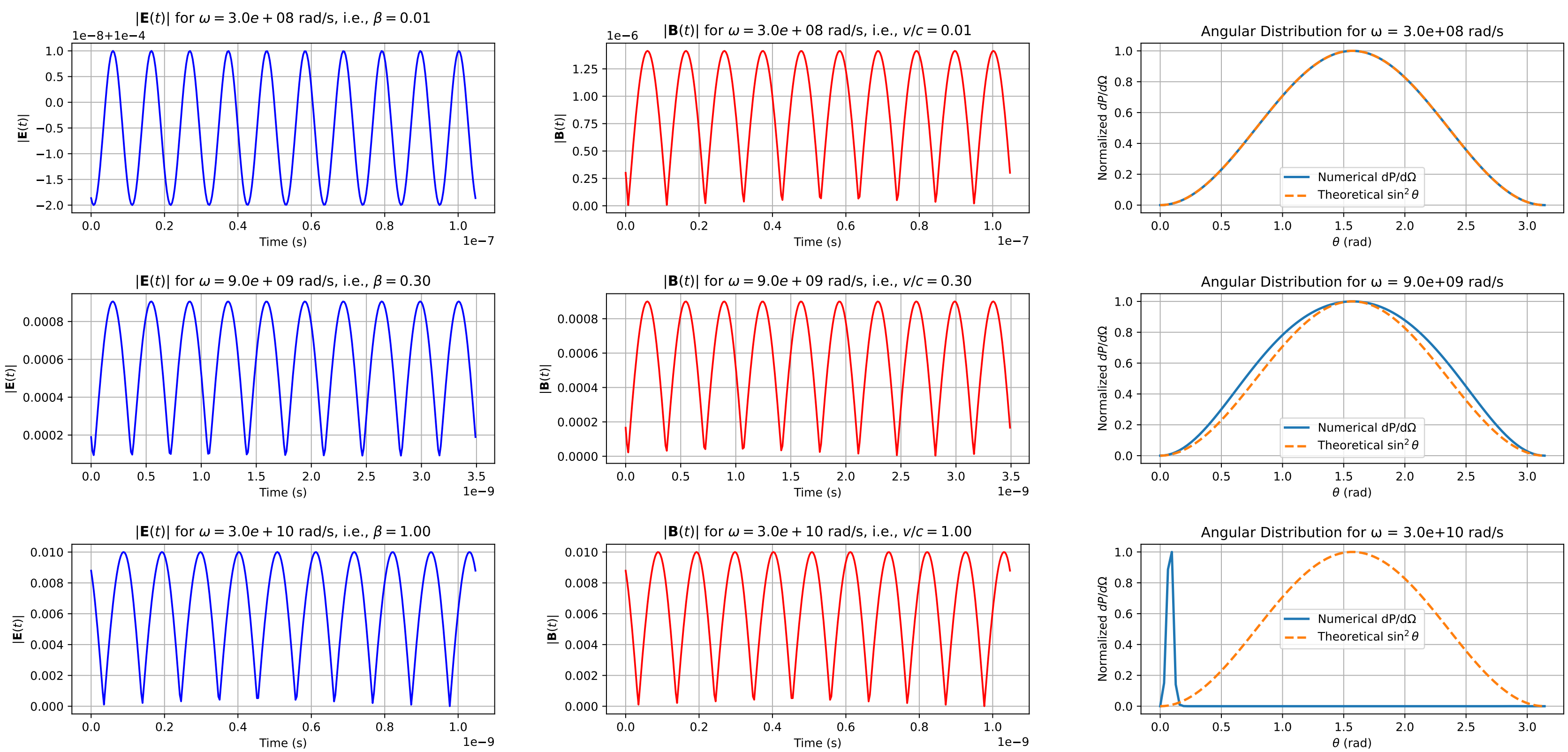


Figure 8: compares the time-evolution of the electric and magnetic field magnitudes, along with the angular distribution of radiated power, for three different source velocities, $\beta = 0.01, 0.3, 1.0$.

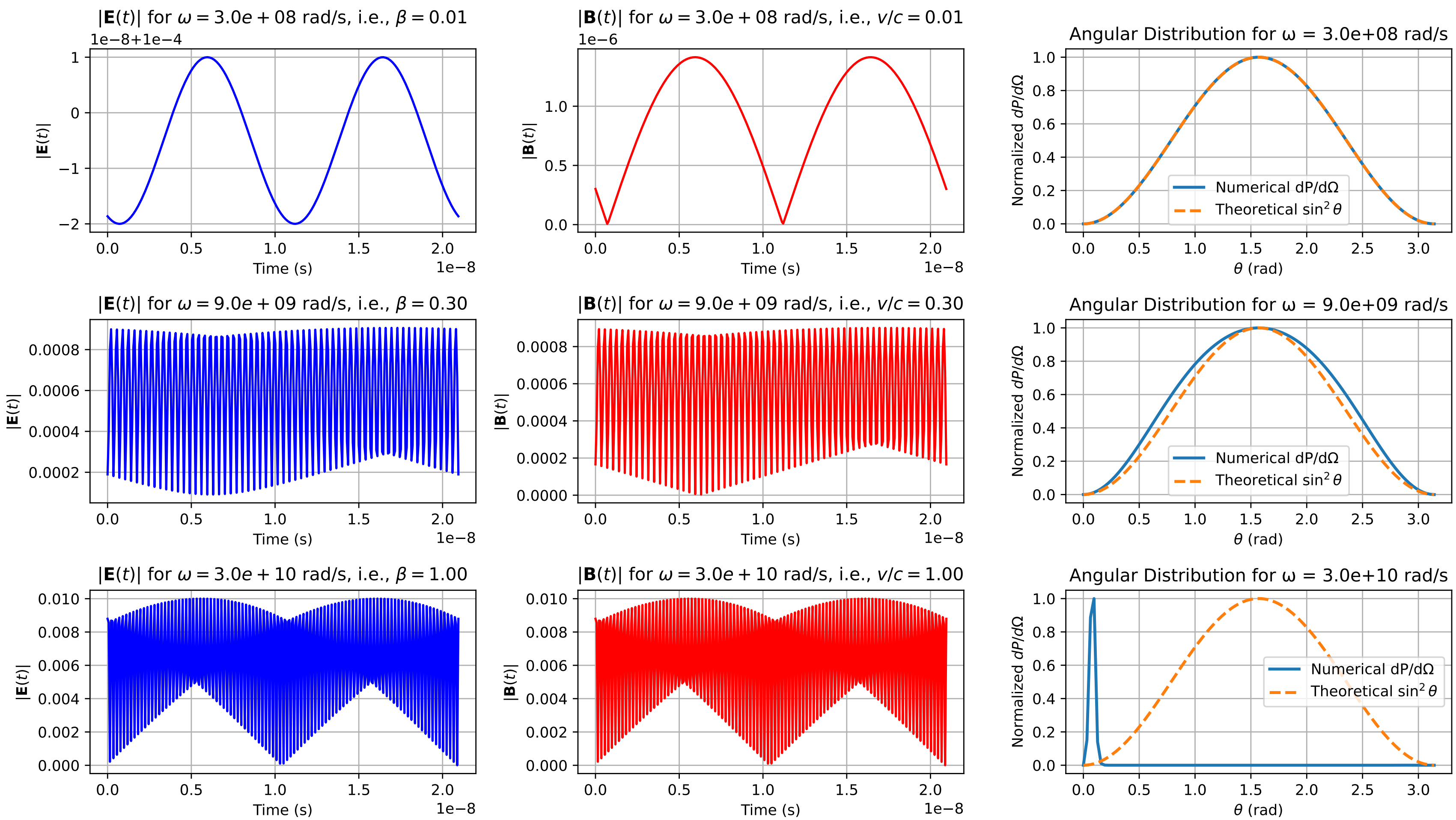


Figure 9: is similar to last figure but with more time period to observe the insanity.

In fig. 8, we can observe that in the Newtonian limit, radiation pattern closely follows $\sin^2\theta$, indicating an isotropic emission profile in the plane orthogonal to the acceleration. As the velocity increases, deviations from this pattern become evident: the radiation becomes increasingly asymmetric and concentrated along the direction of motion - shifting from equatorial dominance toward the poles. This forward beaming effect is characteristic of relativistic motion. The comparison with the $\sin^2\theta$ curve highlights the transition from classical to relativistic radiation characteristics.

Now, repeating the same for circular motion.

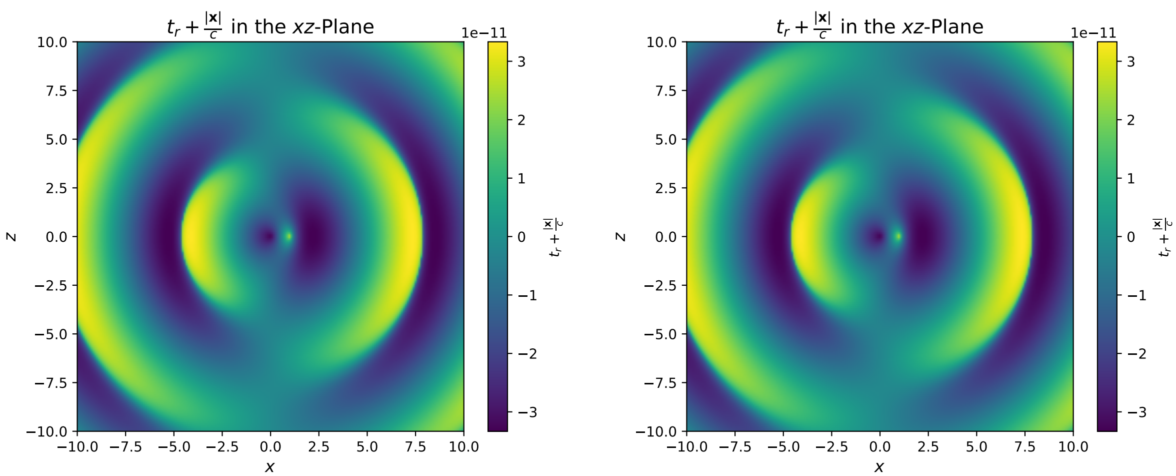


Figure 10: The symmetry of circular motion (in xy plane) is very evident in the plots. The gradient reflects how the light-cone condition modifies the apparent time structure perceived by observers at different locations.

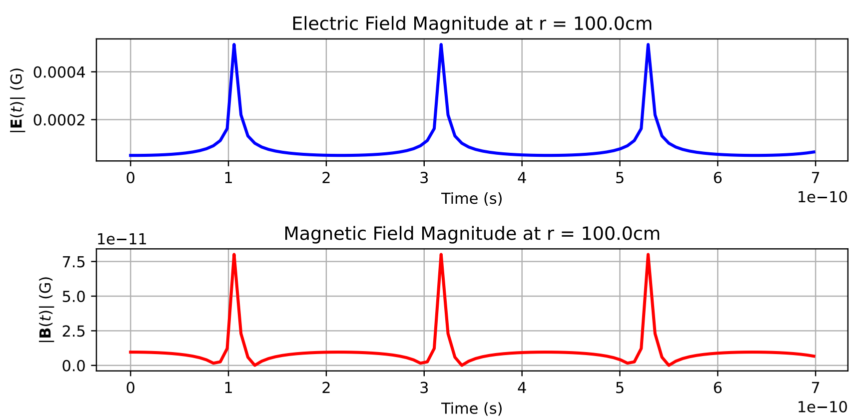


Figure 11: We see $|\mathbf{E}(t)|$ and $|\mathbf{B}(t)|$ are no longer sinusoidal.

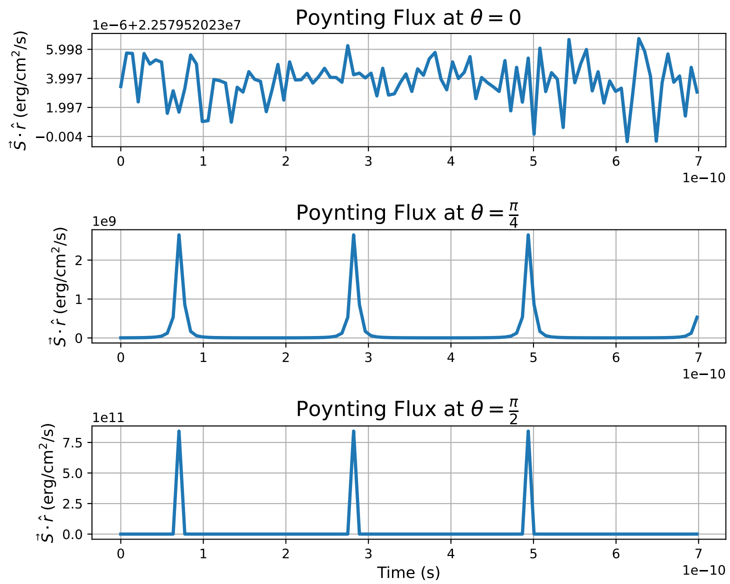


Figure 12: At $\theta = 0$, we can see the flux is almost zero and could be just noise. As θ increases, flux begins to show noticeable structure and indicates the presence of off-axis beaming. This aligns with the expected forward-beaming nature of synchrotron radiation concentrated in the plane perpendicular to acceleration.

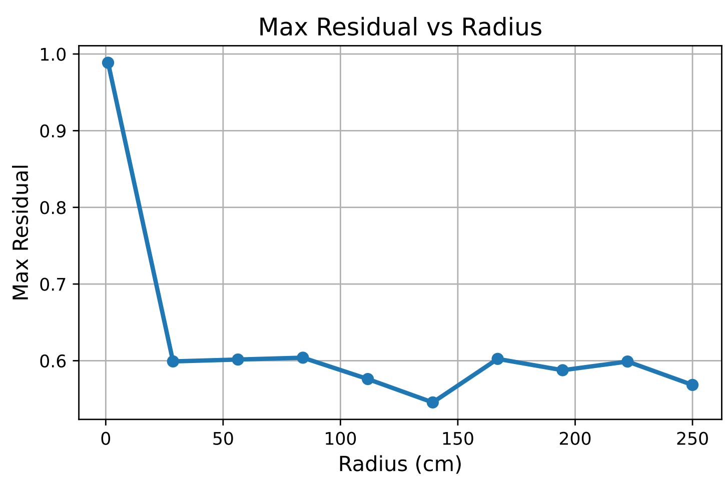


Figure 13: The plot shows far-field starts early here but it is not quite converging.

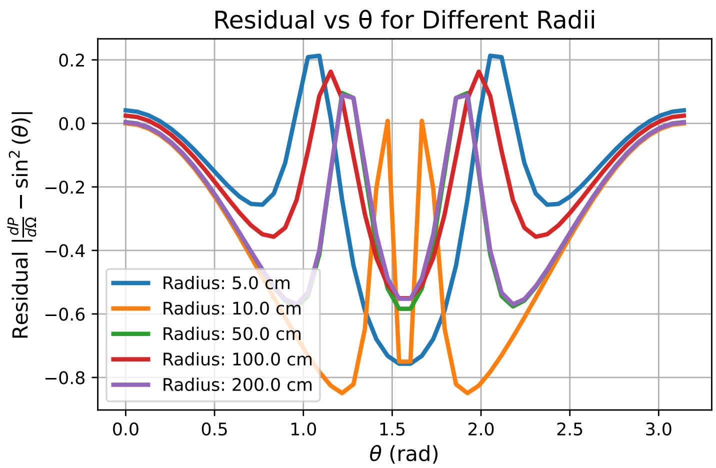


Figure 14: This beautiful graph clearly tells us that we cannot make make the Newtonian limit assumption that there will be $\sin^2\theta$ dependence.

In fig.15, we see for low β, $|\mathbf{E}|$ is sinusoidal and probably $|\mathbf{B}|$ too (my code is faulty:)). However, normalized $\frac{dP}{d\Omega}$ looks weird, like inverted. So , vaguely, we can conclude that there is a $\sin^2\theta$ dependence, I guess. As ω increases, relativistic effects become prominent, resulting in the radiation being beamed forward — shifting from equatorial dominance towards polar dominance. This relativistic beaming is evident in both the field magnitudes (which become sharper and stronger) and the narrowing of the angular power distribution.

I can conclude that my function to find and plot angular power distribution is definitely not working properly for circular ones. And function to calculate $B(t)$ is also not working fine :(

3 Conclusion

In this project, we explored electromagnetic radiation from an accelerating point charge undergoing both linear oscillatory and circular (synchrotron-like) motion. Using the Liénard–Wiechert potentials as a foundation, we implemented a numerical solver that accurately computes the retarded time, electric and magnetic fields, Poynting flux, and angular power distributions. Through systematic validation, we confirmed expected behaviors in the Newtonian regime—such as the $\sin^2\theta$ angular dependence and r^2 energy fall-off. In the relativistic regime, we observed the forward-beaming effect and deviations from classical patterns, aligning with known features of synchrotron radiation.

We can extend the code to involve spectral analyis of the power, accurate $\mathbf{B}(t)$ fields, relativistic corrections(?), better root solvers, etc.,.

The code for these beautiful graphs can be found here:)

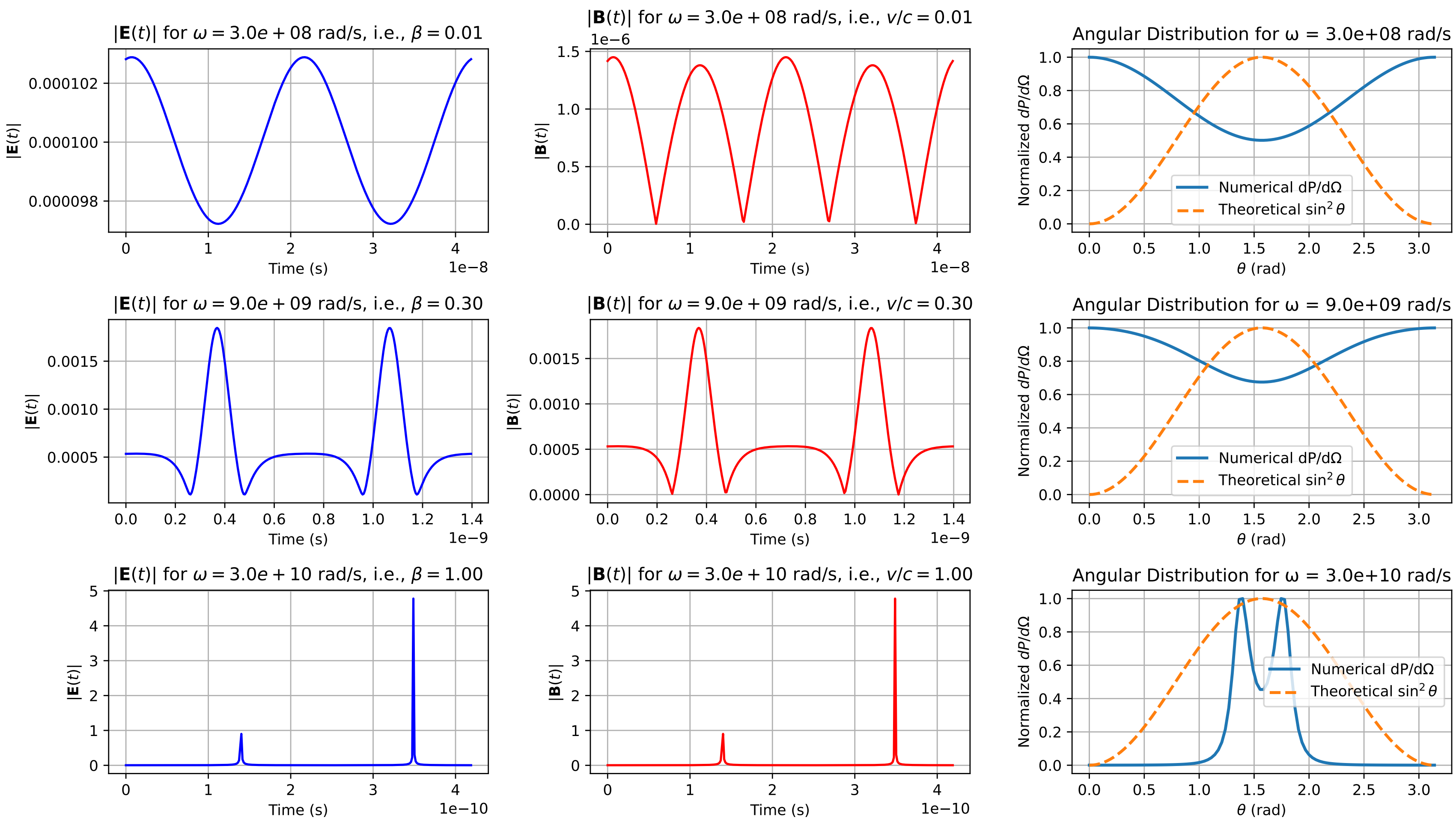


Figure 15: Time evolution and angular distribution of radiation for a charge in circular motion at different velocities.